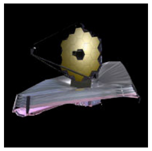

7119 - Does Reionization Quench All Ultra-faint Dwarf Galaxies? The Ancient Star Formation History of Pegasus W

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Observation Folder				
	1		NIRCam Imaging	(1) Pegasus-W

ABSTRACT

We request F090W+F150W NIRCam imaging of Pegasus W (M_V=-7.2; D=915 kpc), an isolated ultra-faint dwarf galaxy on the far side of the Local Group and beyond the virial radius of M31 itself. Pegasus W is a unique ultra-faint, as HST imaging indicates that it quenched 7.4 Gyr ago, far more recently than the ultra-faint satellites of the Milky Way, and, remarkably, that its star formation may have re-ignited briefly ~500 Myr ago.

Reionization is expected to quench the smallest galaxies at early times (>10 Gyr), heating their gas reservoirs and stopping any subsequent star formation. However, the effects of reionization may vary based on the mass of the dwarf, and on the local ionizing field, neither of which has been explored. Pegasus W is the first isolated ultra-faint dwarf which may have been resistant to the effects of reionization, but resolved stellar population observations down to the oldest main sequence turnoff are necessary to robustly verify its ancient star formation history. Only JWST+NIRCam can perform these observations in a reasonable amount of time at the distance of Pegasus W, allowing us to probe reionization's role in the smallest dark matter halos in a new regime.

We have designed our observations with future proper motion measurements in mind to further understand the dynamical history of this unique ultra-faint dwarf. We also waive proprietary rights so
the full community can immediately utilize this potential touchstone data set.

OBSERVING DESCRIPTION

We will image Pegasus W with NIRCam with the short-wavelength filters F090W and F150W for our primary science goals, while simultaneously obtaining F277W data. Pegasus W has a small size on the sky and fits in one NIRCam detector. We will position it on one of the module B detectors (using a postarg-offset). We will use both NIRCam modules to fully characterize Pegasus W and the background.

We will image down to F090W=28.9, F150W=28.4 with a S/N=5, assuming a G5V star spectrum, similar to that expected for main sequence turnoff stars. We have used a DEEP8 readout and subpixel dither pattern. We will not cover the detector gaps, and prefer the uniform depth for our science goals. All exposure times have been calculated using the JWST ETC. Pegasus W has two windows of visibility of ~3 weeks each.

We are happy to work with STScI staff if this or any other aspects of our program can be made more efficient.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Pegasus-W	RA: 23 53 14.9952 (358.3124800d) Dec: +22 06 7.09 (22.10197d) Equinox: J2000		

Comments: This object was generated by the targetselector and retrieved from the NED database.
Category=Galaxy
Description=[Dwarf galaxies]

Observation

Proposal 7119, Observation 1

Mon May 19 23:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

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Comments: This object was generated by the targetselector and retrieved from the NED database.
Category=Galaxy
Description=[Dwarf galaxies]

Template

Module	Subarray	Target Placement
ALL	FULL	Module Gap

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		SMALL-GRID-DITHER		8

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F090W	F277W	MEDIUMDEEP8	5	2	16	8	11767.5	220364
2	F150W	F277W	MEDIUMDEEP8	5	2	16	8	11767.5	220364

Special Requirements

Offset 114.40002561171426 arcsec, 31.06162348707371 arcsec